Kerala Home, Phase 1

Pre-construction drawing set, Guntur, Andhra Pradesh (agricultural land: conversion to be confirmed)

Sheet index

A-000	Cover and sheet index	-
A-001	Design basis and general notes	-
A-101	Site plan and services	1:100
A-102	Ground floor plan	1:75
A-103	Furniture layout	1:75
A-104	Working plan: clear sizes and sections	1:75
A-105	Floor finishes and levels	1:75
A-106	Roof plan	1:75
A-201	North and south elevations	1:75
A-202	East and west elevations	1:75
A-301	Sections A-A and B-B	1:75
A-401	Door and window details	1:60
A-501	Typical details 1: plinth, floor, window, eaves	NTS
A-502	Typical details 2: bath, kitchen, basin, AAC	NTS
A-503	Typical details 3: dining ventilator	NTS
A-601	Schedules	-
E-101	Lighting, fans and ceilings	1:75
E-102	Power, AC and data	1:75
P-101	Water supply and drainage	1:100

Project data

Site	Guntur, Andhra Pradesh; agricultural land (conversion to confirm)
Plot	27' x 75' = 2,025 sq ft, + 10' north strip owned (Phase 2)
Built footprint	60' x 22' = 1,320 sq ft (wall centrelines)
Carpet area	1,200 sq ft clear, incl. poomukham and open wet area
Construction	RCC frame, AAC walls, steel-truss roof with clay tiles
Plinth	FFL +2'-0" above finished ground

Key design decisions

- Site is in Guntur, Andhra Pradesh (farm land), not Kerala; Kerala style kept, construction adapted to local climate and soil.
- Plot 27 ft x 75 ft (5 ft longer than first assumed; confirm by survey), road on east; house 60 ft x 22 ft with the same 10 ft front yard and 5 ft rear garden; owner accepts local practice on side setbacks (to be checked).
- RCC frame with AAC block walls: 9 in outer, 5 in inner, 7 in where plumbing is concealed.
- Plinth 2 ft above finished ground; poomukham 6 in lower; bathrooms 12 mm lower with no raised threshold.
- AC provisions (points, sleeves, drain, outdoor-unit brackets) in both bedrooms, the office and living; passive cooling first.
- Flat insulated false ceilings in bedrooms, offices, baths and kitchen; sloped ceilings in living, dining and verandah.
- No glass skylight: the dining gets borrowed daylight from the living, the nook window and the hall; a louvred ventilator lantern above it lets hot air out (no glass, so no greenhouse heat).
- Courtyard removed; the dining hall at plot centre keeps the Brahmasthan open, with the ventilator above.
- Single storey, no internal stair; roof access by fixed ladder; water tank on stand in SW rear garden.
- Service band on south (heat buffer); bedrooms and living get north light.
- Kitchen in SE; stove on east wall; open wet area onto the walkway; provisions in bench drawers, kitchen cabinets and lofts (no separate store).
- Bedroom doors widened to 3 ft; bathroom, office and store doors 2 ft 6 in.
- Eighth review: plot 5 ft longer, so the house grows to 60 ft. The master suite takes 2 ft (master bedroom 11'-6" x 14', master bay 9 x 8 ft) and Bedroom 2 takes 3 ft (15 x 16 ft); the sitting area in front of them grows to 18 ft. Everything east of Bedroom 2 moves 5 ft east; the plot centre now falls inside Bedroom 2, near its east wall, with the dining and ventilator about 5 ft east of it (Vastu to confirm).
- Seventh review: Bedroom 2 enlarged to 12 x 16 ft by taking the hall strip; its door opens from the sitting area (as the master suite does); hand-wash basin back at the west end of the serving counter.
- Sixth review: Office 2 walls and sliding door left out for now (open workstation in the master suite, walls can be added later); Bedroom 2 becomes a study-guest room: work table with monitor beside the bath, sofa-cum-bed instead of a queen bed.
- Office 1 removed after family review: one office (Office 2) inside the master suite; Bedroom 2 has its own work desk.
- Common bathroom moved to the north side (old pooja spot), entered from the dining corner next to the hall.
- Second family review: store (8 x 8 ft) moved next to the bedrooms; pooja reduced to a small shrine opening onto the hall (idols face east); wet area moved west (9 x 8 ft); kitchen 14 x 10 ft with a 5 ft sliding entry from the living; dining gains a basin and crockery nook with a south window; TV moved to the living room's east wall.
- Pooja moved beside the wet area and opened to the west (idols face west, the family sits in the nook facing east); shared wall kept dry with no pipes, pooja floor 3 in higher; basin nook shifted west.
- Fifth review: Bedroom 2 made one 12 x 12 ft rectangle with a rectangular private bath (north-west corner, outer wall) and its wardrobe on the east wall; the leftover corner at the hall's west end is an open hand-wash corner; serving counter now 9 ft without a basin.
- Fourth review: Bath 2 made Bedroom 2's private bath (no common toilet); master suite entered through its bay from the sitting area; the old Bath 2 spot opens to the dining with a north window; family store merged back into the master bedroom as a wardrobe bay; inverter under the sitting-area bench.
- Earlier: walk-in wardrobe turned into a family store with a lockable door from the sitting area (inverter inside); built-in sliding wardrobes in the dressing alcove and on the master bedroom's south wall.
- Store wall removed: the strip from the walk-in to the pooja is one open sitting and serving area with a storage bench; hand-wash basin set into the west end of the serving counter, about 8 ft from the pooja, draining straight out to the south walkway.
- Wet area reduced to 6 x 6 ft and left open on its south side to the walkway (service route from the kitchen to the back of the house); kitchen door moved to face the dining (3 ft sliding glass); basin nook widened to 8 ft with a 6 ft serving counter.
- Main door moved to the south part of the living's east wall (still north of the house's centre line); TV on the north part of that wall; steps, swing and arugu seats rearranged on the poomukham.
- Third review: hall widened to about 6 ft clear (store, pooja and basin nook made 6 ft deep) so the family can sit facing the pooja; store left open to the hall for now (a wall and door can be added later); broom closet dropped, tall utility cupboard in the wet area instead.

Kerala Home, Phase 1

Kerala-style single-storey residence
Guntur, Andhra Pradesh
Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Cover and sheet index**
Sheet	**A-000**
Scale	-
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Site conditions (Guntur)

• **Heat:** summer days 40-45 C (March-June). Insulated roof, ventilated attic, shaded west and south walls, AC provisions.

• **Rain:** about 850-1000 mm a year, mostly June-October, with short cyclonic downpours in Oct-Nov. Deep eaves, gutters, drip grooves.

• **Wind:** coastal cyclone belt. Trusses bolted down; every edge tile clipped; no light sheeting that can lift.

• **Soil:** black cotton (expansive) soil is common. A soil test decides the foundation; keep water and trees away from the walls.

• **Earthquake:** seismic zone III. RCC frame with lintel and ring bands.

• **Water:** groundwater is often hard with high fluoride. RO for drinking, softener if needed, matt fittings.

• **Termites and mosquitoes:** anti-termite treatment before the plinth; mesh on every opening.

• **Land:** agricultural land. Confirm conversion and the permission route before work starts.

Levels and heights

Level	Above finished ground
Finished ground (FGL)	+0'-0"
Poomukham floor	+1'-6"
Wet area floor	+1'-9"
House floor (FFL)	+2'-0"
Bathroom floors	+1'-11.5" (12 mm drop)
Door head / lintel band	+9'-0"
False ceilings (rooms)	+11'-6"
False ceilings (baths)	+10'-6"
Wall top / ring beam	+12'-0"
Roof eaves (tile edge)	+10'-6.5"
Main ridge	+18'-4"
Front gable ridge	+15'-1.5"

Wall types

Type	Build-up	Where
EXT	200 mm AAC + 20 mm external plaster + 12 mm internal plaster (~9 in)	Outer walls, kitchen north wall
WET	150 mm AAC + plaster both faces (~7 in): concealed plumbing walls	Walls with concealed plumbing
INT	100 mm AAC + plaster both faces (~5 in)	Other inner walls

General notes

• Dimensions are to wall centrelines unless marked 'clear'. Clear room sizes are on A-104.

• Concrete, steel, foundations and member sizes are by the structural engineer. Column positions are not shown and must be coordinated with the furniture layout.

• AAC: 150 mm galvanised mesh over every block-concrete joint; chases cut by machine; no hammering.

• All external sills, sunshades and copings get drip grooves; every external opening gets mosquito mesh.

• Wet rooms: waterproofing on floors, 12 in up the walls and 7 ft in showers; 48 h flood test before tiling.

• Skirting is flush with the plaster (no dust ledge). Grout in wet areas is epoxy.

• Pipes and conduits are laid before plaster and photographed with a tape measure in view, for future drilling.

• Keep 3 ft paved apron around the house, sloped away. No gardens, soak pits or big trees within 6-8 ft of walls.

Hold points (stop and inspect)

Stage	Check before it is covered up	Who
Before design freeze	Soil test report, level survey of plot and road, water quality test (TDS, hardness, fluoride)	Owner, engineer
Excavation	Footing depth and bearing layer seen in the pit; no black soil left under footings	Structural engineer
Before each concrete pour	Reinforcement, cover blocks, laps and column positions checked; photos taken	Engineer / site supervisor
Plinth	Anti-termite treatment (certificate), fill compaction, DPC laid and cured	Supervisor
Before plaster	Conduits, pipes, AC sleeves, mesh over AAC joints; photos with a tape measure in view	Owner, supervisor
Wet rooms	Waterproofing coats, upturns, 48 h flood test passed before any tile is laid	Owner, plumber
Roof	Truss holding-down bolts, membrane and insulation, edge tiles clipped, flashings at the ventilator and valleys	Engineer, owner
Before handover	Water test on every floor slope and sill, plumbing pressure test, electrical insulation and earth test	Owner, MEP

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

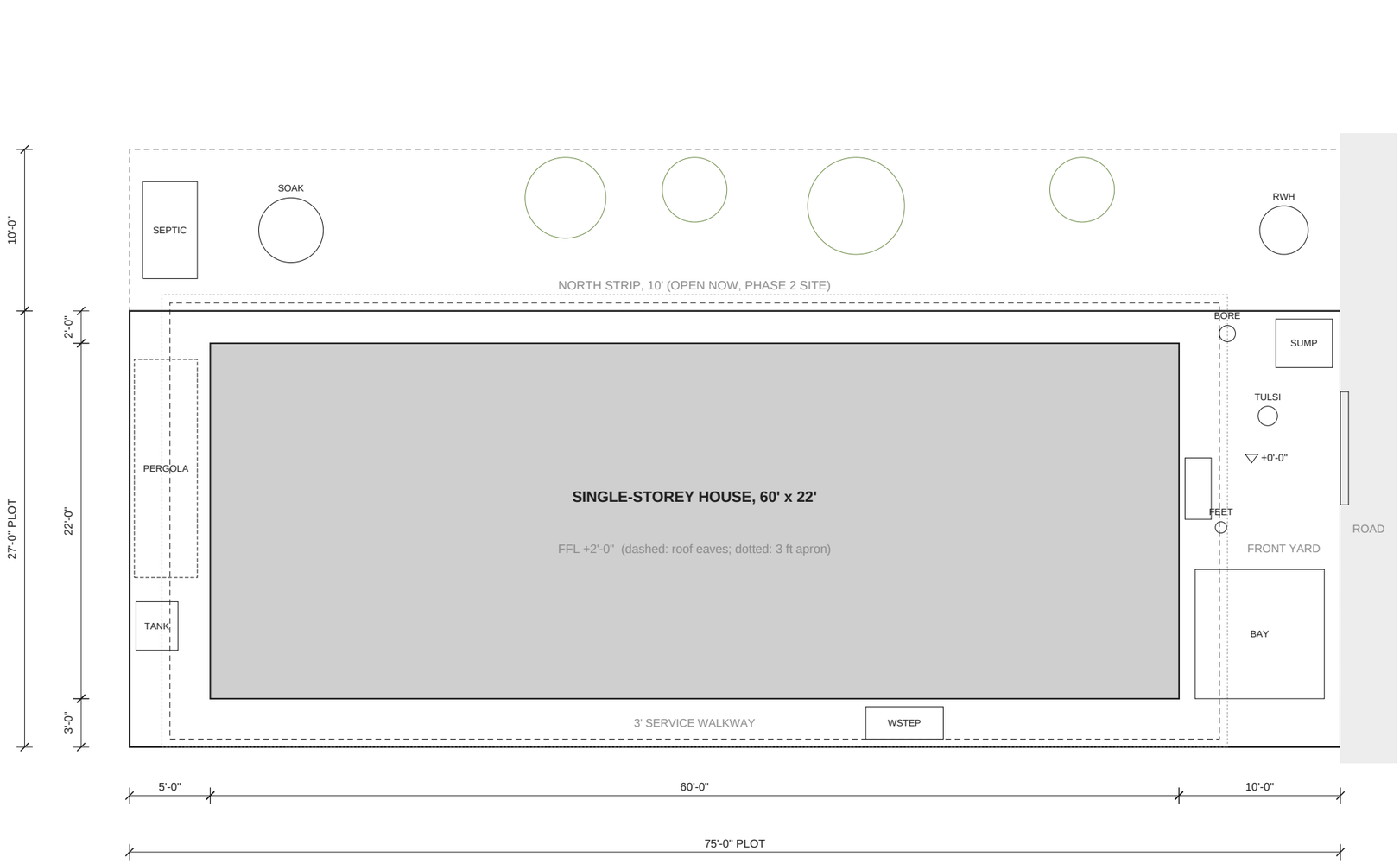
Stage	Draft for professional review
Drawing	**Design basis and general notes**
Sheet	**A-001**
Scale	-
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east



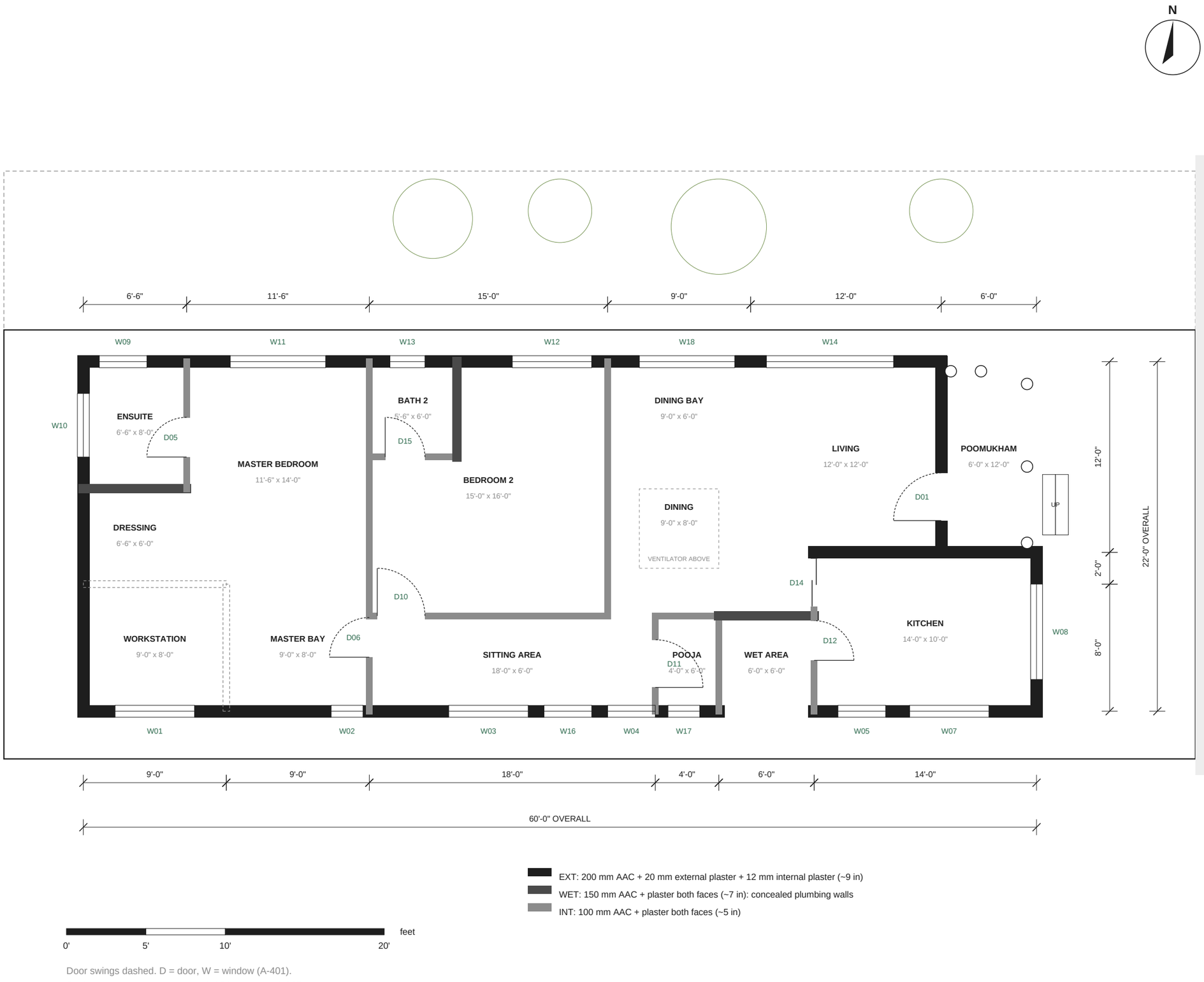
- TANK: Overhead tank on 18 ft stand (SW), ladder to roof
- SEPTIC: Septic tank, 2 chambers (NW, in north strip)
- SOAK: Soak pit, grey water (6 ft+ from house)
- SUMP: Sump (NE)
- BORE: Borewell (if any), NE
- RWH: Rainwater recharge pit (sump overflow)
- PERGOLA: Creeper pergola for west shade (no big trees near walls)
- TULSI: Tulsi thara
- BAY: Two-wheeler bay
- GATE: Gate
- STEPS: Entrance steps (3 risers), between the pillars
- FEET: Foot-wash tap + drain
- WSTEP: Steps from the open wet area down to the walkway (4 risers of about 5 in, along the wall)
- APRON: Plinth protection apron 3 ft, sloped away
- Site drains: finished ground falls 1:100 away from the house to the road side.

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Site plan and services**
Sheet	**A-101**
Scale	1:100 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Kerala Home, Phase 1

Kerala-style single-storey residence
Guntur, Andhra Pradesh
Plot 27' x 75', road on east



Door swings dashed. D = door, W = window (A-401).
Section lines and clear sizes on A-104; levels on A-105.

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

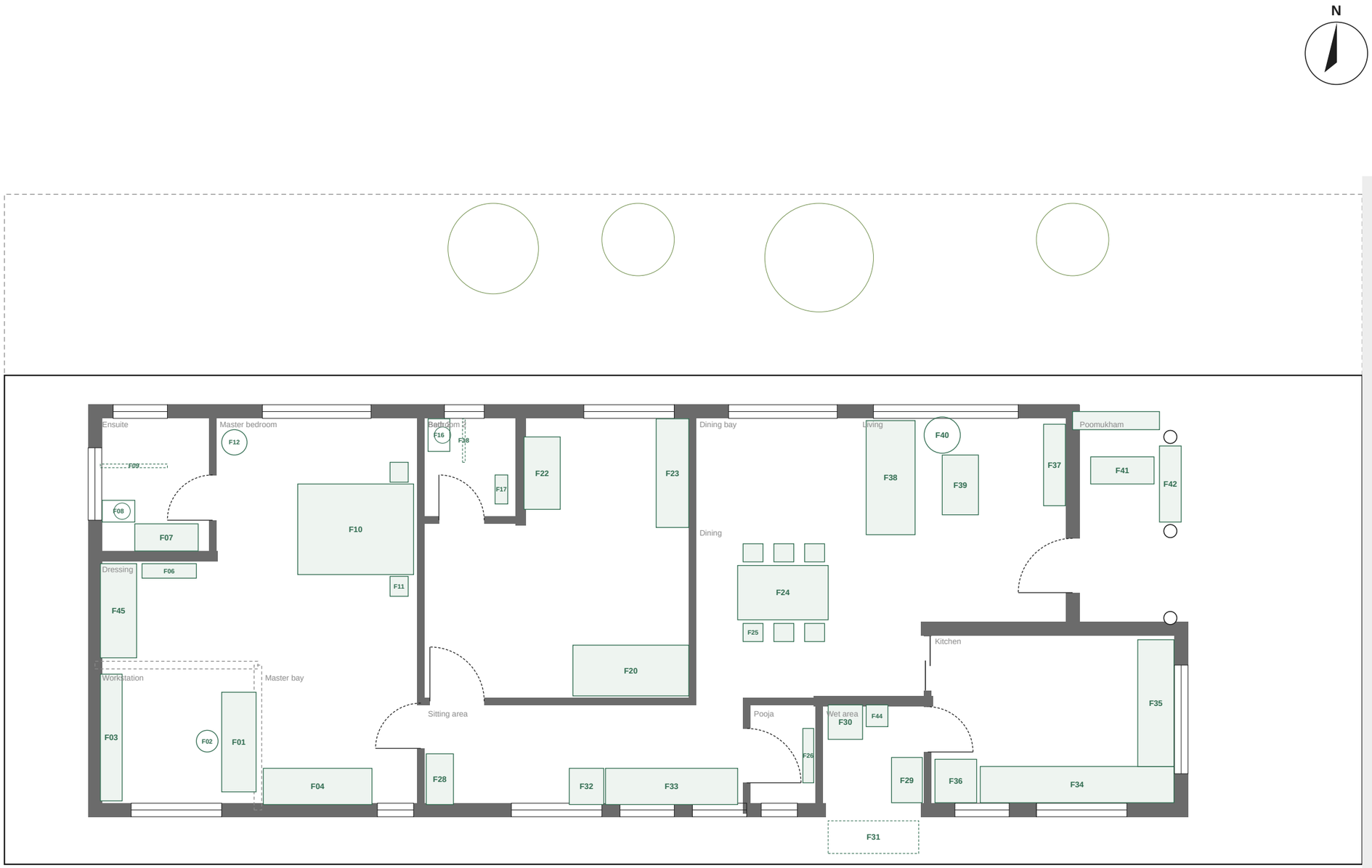
Stage	Draft for professional review
Drawing	**Ground floor plan**
Sheet	**A-102**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east

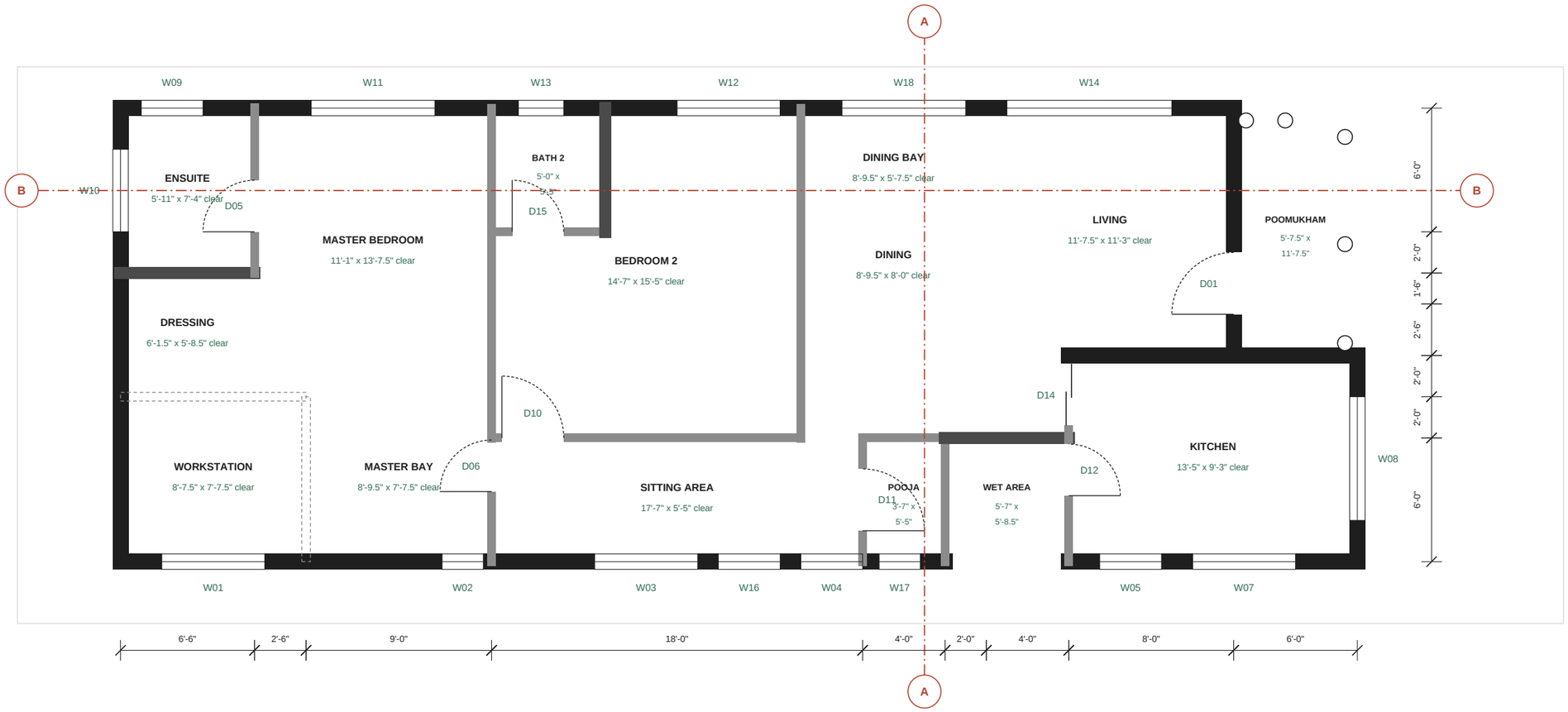


Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Furniture key (full specs in schedules)

- **F01** Work desk 5'6" x 2' (free-standing, cable tray)
- **F02** Ergonomic task chair
- **F03** Bookshelf, full height (video backdrop)
- **F04** Built-in wardrobe 6 ft, south wall, sliding doors, lofts above
- **F06** Dresser with mirror (north wall)
- **F45** Built-in wardrobe, west wall, sliding doors
- **F07** Vanity basin counter 3'-6"
- **F08** WC (EWC)
- **F09** Shower glass partition
- **F10** Queen bed 5' x 6'6" + headboard
- **F11** Bedside table (x2)
- **F12** Reading chair
- **F16** WC (EWC)
- **F17** Wash basin
- **F18** Shower glass partition (shower in the north-east corner)
- **F20** Sofa-cum-bed 6'6" (opens to a 5 ft deep bed)
- **F22** Work table 4' x 2' with monitor (against the bath wall)
- **F23** Built-in wardrobe 6' (east wall, toward the house centre)
- **F24** Dining table 5' x 3', 6-seat
- **F25** Dining chair (x6)
- **F26** Pooja mandapam on the east wall (idols face west; family faces east)
- **F28** Built-in bench with storage drawers below
- **F29** Utility sink counter (on the kitchen wall)
- **F30** Washing machine bay 2' x 2' (covered back corner, 2 in clear of the pooja wall)
- **F31** Fold-away clothesline in the south walkway
- **F44** Tall slim utility cupboard (brooms, mops, detergents)
- **F32** Counter-top wash basin at the west end of the serving counter, away from the pooja
- **F33** Serving counter with crockery cabinets above, 7 ft
- **F34** L-counter, south run
- **F35** L-counter, east run (hob)
- **F36** Refrigerator space (west end, away from the hob)
- **F37** TV unit + 50-55 in TV (east wall, north of the door)
- **F38** Three-seat sofa 6'6" facing the TV
- **F39** Coffee table 3'6" x 2'
- **F40** Armchair by the north window
- **F41** Oonjal (swing) 3'6"
- **F42** Arugu seats (built-in) (x2)

Stage	Draft for professional review
Drawing	**Furniture layout**
Sheet	**A-103**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground



Clear sizes are wall face to wall face after plaster (green). Grid dimensions are to wall centrelines.

Section A-A and B-B on A-301. Check every furniture item against the clear size before ordering.

Column positions (by engineer) must avoid door swings, wardrobes and the TV wall.

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Working plan: clear sizes and sections**
Sheet	**A-104**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east

N



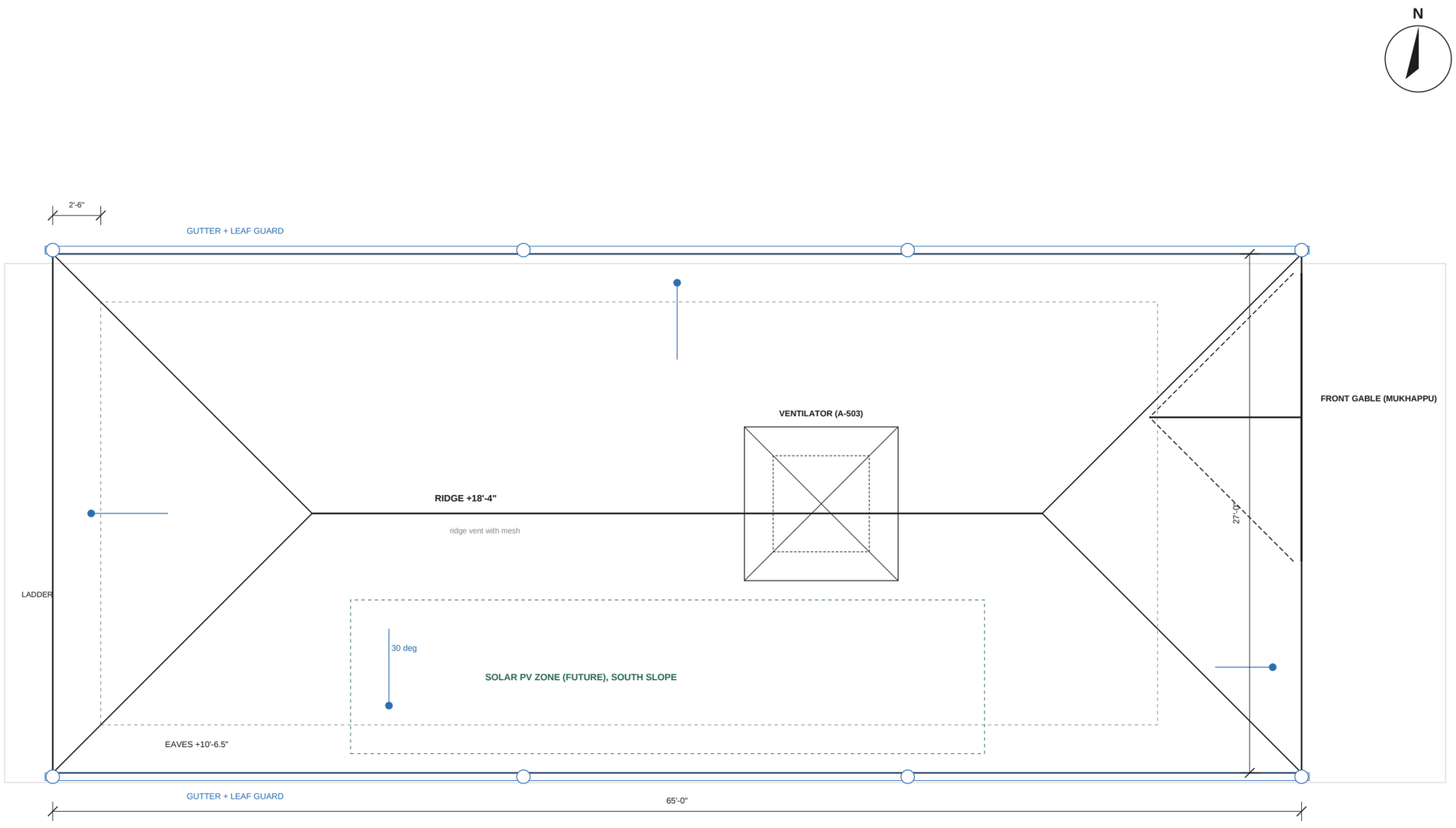
- Red oxide / Kota stone
- Matt vitrified 2x2
- Marble / teak (pooja)
- Anti-skid matt vitrified
- Anti-skid ceramic, waterproofed
- Rough Cuddapah / Kota, falls
- Flamed granite

Floor detailing

- Skirting 4 in high, set flush with the plaster face (plaster stops in a groove above it), so no ledge collects dust.
- Bathrooms 12 mm below the bedroom floor with a sloped granite threshold: no raised step to trip on, water stays in.
- Shower zone: linear drain at the far end, floor falls 1:60 only inside the wet zone; dry zone floor nearly flat.
- Poomukham and steps: flamed or leather-finish stone. Polished granite is slippery when wet.
- Kitchen: mid-tone matt floor; white glossy tile shows every stain and is slippery with oil.
- Leave 5 mm expansion gaps at walls under the skirting and every 15-20 ft in large floors.

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Floor finishes and levels**
Sheet	**A-105**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground



Roof detailing (Guntur: heat, cyclone wind, short heavy rain)

- Every tile along eaves, verges, hips and ridges clipped or screwed; every third row elsewhere. Cyclone winds lift loose tiles first at the edges.
- Trusses bolted to the RCC ring beam with holding-down bolts; purlins cleated, not just welded tacks.
- Eaves 2'-6" beyond the wall line with a drip edge so water falls clear of the wall and windows.
- Gutters at both long eaves, downpipes at 8 points into the sump filter and recharge pit; none discharging onto bare soil by the walls.
- Ventilator: hipped tiled cap, 45 deg louvres with mesh, sealed inner shutters, 9 in upstand with flashing (detail on A-503).
- Valleys at the front gable: 600 mm wide GI / aluminium valley gutter under the tiles.
- Attic vented at the ridge and soffits with insect mesh; 50 mm insulation on the deck.
- Leave a 25 mm conduit from roof to DB now for future solar PV on the south slope.

Kerala Home, Phase 1

Kerala-style single-storey residence

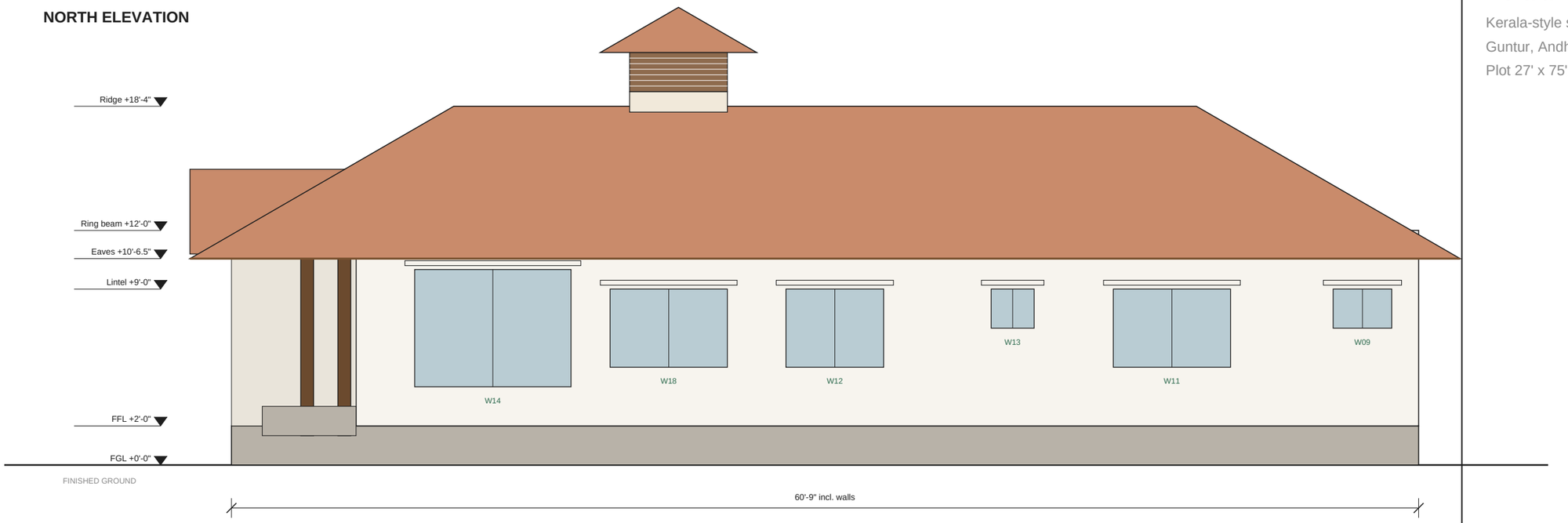
Guntur, Andhra Pradesh

Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Roof plan**
Sheet	**A-106**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

NORTH ELEVATION



SOUTH ELEVATION



Brown line at the eaves = fascia with gutter. Every window has a sunshade (chajja) with a drip groove; sills slope out 1:10.

Kerala Home, Phase 1

Kerala-style single-storey residence

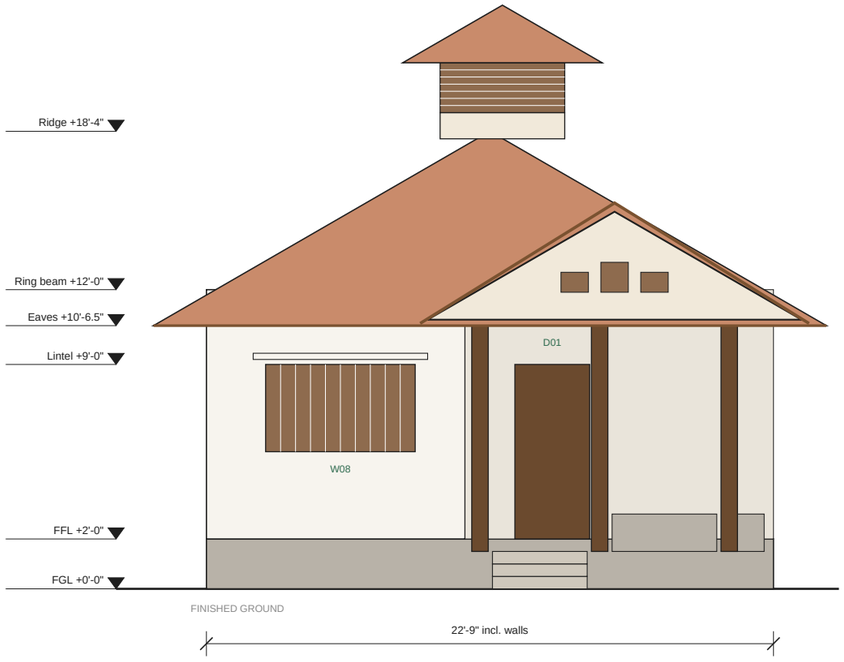
Guntur, Andhra Pradesh

Plot 27' x 75', road on east

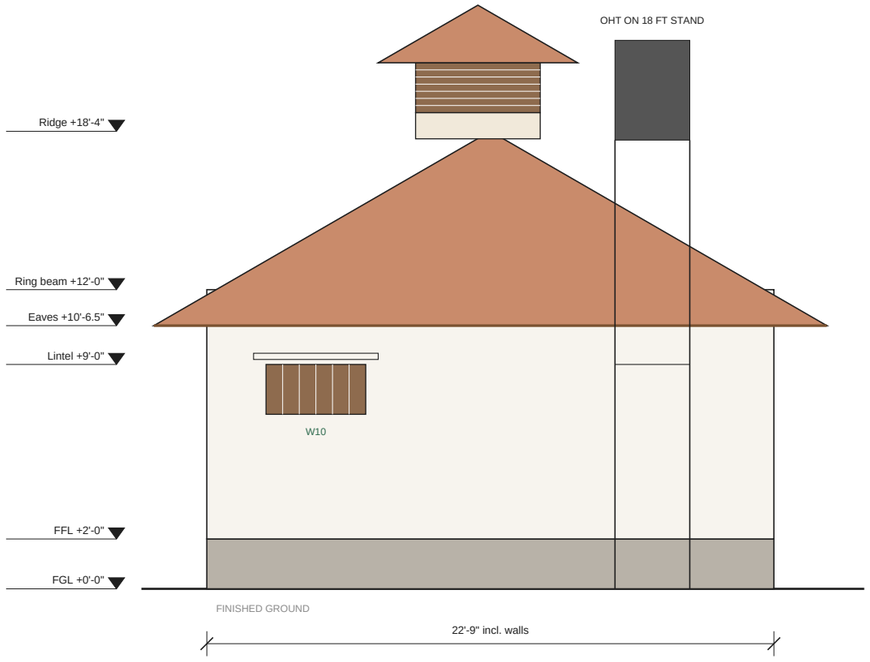
Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**North and south elevations**
Sheet	**A-201**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

EAST ELEVATION (ROAD SIDE)



WEST ELEVATION



- Front gable (mukhappu): three timber jali vents let morning light in and hot air out; insect mesh behind.
- Plinth face clad 2 ft in rough stone: the splash zone stays clean and does not erode.
- West wall: no large windows (afternoon sun). Shade with the creeper pergola, not trees, on black cotton soil.
- Entrance: 3 risers of 6 in, 12 in treads, flamed granite with a nosing; foot-wash tap beside the steps.

Kerala Home, Phase 1

Kerala-style single-storey residence

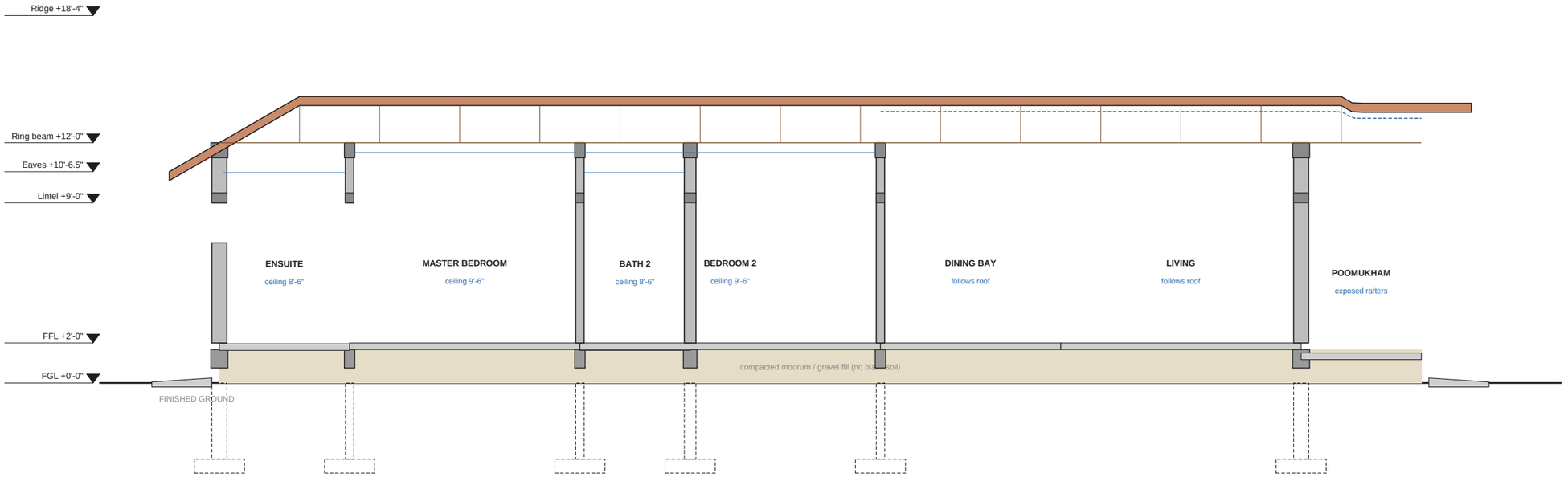
Guntur, Andhra Pradesh

Plot 27' x 75', road on east

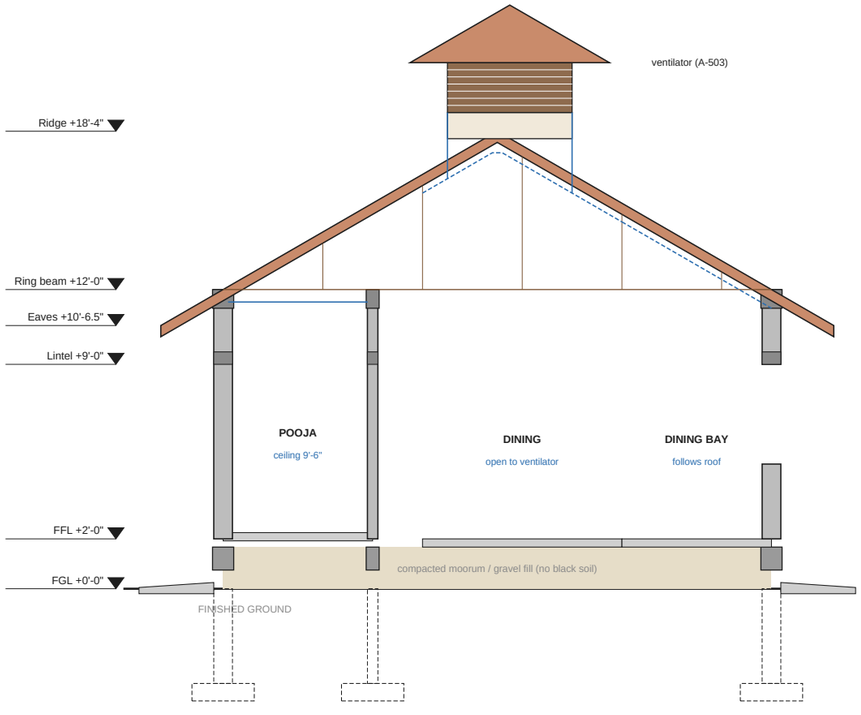
Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**East (road) and west elevations**
Sheet	**A-202**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

SECTION B-B: LONG SECTION THROUGH BEDROOMS, LIVING AND POOMUKHAM



SECTION A-A: CROSS SECTION THROUGH POOJA, DINING AND DINING BAY



Section notes

- Clay tiles on battens, membrane, 50 mm insulation, cement-fibre deck on purlins (attic ventilated at ridge and soffit).
- Steel truss bolted to the RCC ring beam with holding-down bolts (cyclone uplift).
- RCC lintel band at door-head level runs continuously (seismic zone III).
- AAC walls in an RCC frame; mesh over every block-to-concrete joint before plaster.
- Flat false ceilings (blue) in bedrooms, offices, baths and kitchen, with insulation above. Living and dining follow the roof.
- Plinth beam with DPC; plinth fill of compacted moorum, never black soil; anti-termite treatment at every layer.
- Footings dashed and indicative only: type and depth by the engineer after the soil test.
- 3 ft PCC apron sloped away from the wall; joint against the plinth sealed.

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

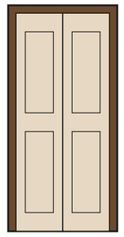
Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Sections A-A and B-B**
Sheet	**A-301**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

DOORS (viewed from the side the door swings into; vents above inner doors, 7'-6" to 9'-0"; none at bedrooms or bathrooms)

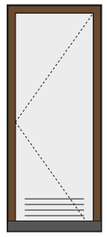
D01



3'-0" x 7'-0"

Teak, double leaf, panelled; 4 in raised threshold outside slope; tower bolts + mortise lock

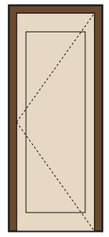
D05



2'-6" x 7'-0"

WPC/FRP shutter, frame on granite base; louvre at bottom
No vent above

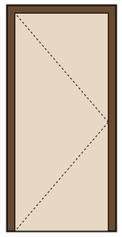
D06



2'-6" x 7'-0"

Teak panelled, single leaf
No vent above

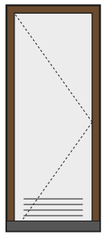
D10



3'-0" x 7'-0"

Flush, teak veneer
No vent above

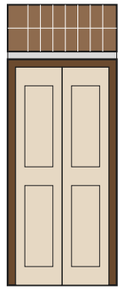
D15



2'-6" x 7'-0"

Flush, waterproof (WPC/FRP)
No vent above

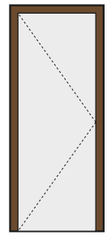
D11



3'-0" x 7'-0"

Carved teak, double leaf (opens west; family sits facing east)
Above: open jali

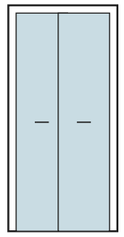
D12



2'-6" x 7'-0"

WPC/FRP shutter, frame on granite base
No vent above

D14

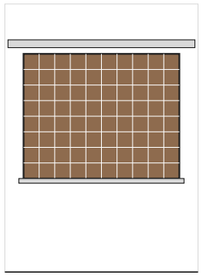


3'-0" x 7'-0"

Sliding glass, 3 ft (closes off cooking smoke)
No vent above

WINDOWS (viewed from inside; sill and head heights above floor)

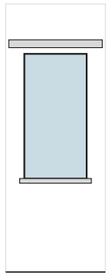
W01



5'-0" wide, sill 3'-0", head 7'-0"

Timber jali + glazed shutters

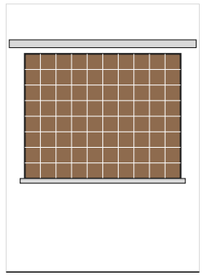
W02



2'-0" wide, sill 3'-0", head 7'-0"

Teak casement, glazed (south light, deep sunshade)

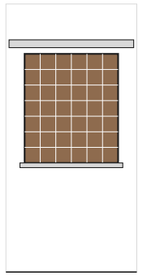
W03



5'-0" wide, sill 3'-0", head 7'-0"

Timber jali + glazed shutters

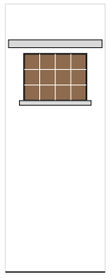
W04



3'-0" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters (over the serving counter)

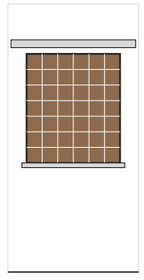
W17



2'-0" wide, sill 5'-6", head 7'-0"

High vent, jali + mesh (lamp smoke out)

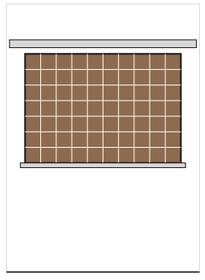
W05



3'-0" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters

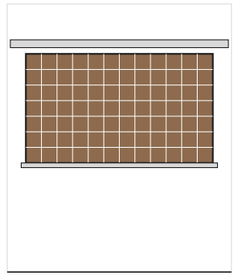
W07



5'-0" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters

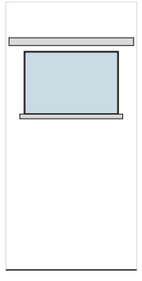
W08



6'-0" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters

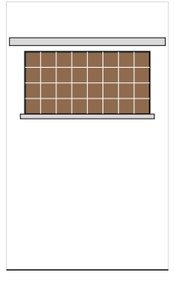
W09



3'-0" wide, sill 5'-0", head 7'-0"

Frosted casement

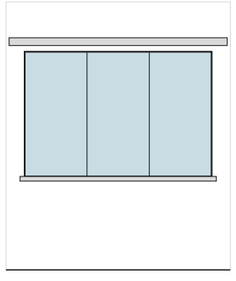
W10



4'-0" wide, sill 5'-0", head 7'-0"

High jali vent

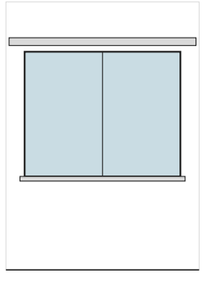
W11



6'-0" wide, sill 3'-0", head 7'-0"

Teak casement, glazed

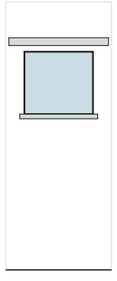
W12



5'-0" wide, sill 3'-0", head 7'-0"

Teak casement, glazed

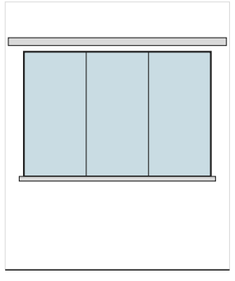
W13



2'-2.5" wide, sill 5'-0", head 7'-0"

Frosted casement, top-hung vent

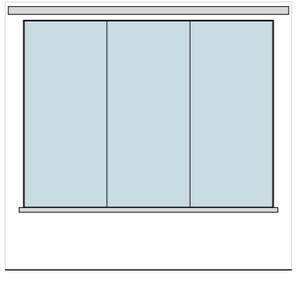
W18



6'-0" wide, sill 3'-0", head 7'-0"

Teak casement, glazed (north light for the dining)

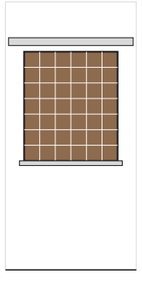
W14



8'-0" wide, sill 2'-0", head 8'-0"

Teak casement, tall, glazed

W16



3'-0" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters (over the serving counter)

- All windows: insect mesh (SS or fibreglass) on a separate openable or sliding frame, so glass can be cleaned.
- MS safety grill inside the shutters on every ground-floor window; one escape window per bedroom with an openable grill section.
- Teak is costly and must be seasoned; good alternatives in Guntur: sal or hard pinkado for frames, UPVC for wet-side windows.
- Door frames 5 x 2.5 in; frames in bathrooms and wet areas stand on granite or WPC blocks, never touching the wet floor.
- Hardware: SS 304 hinges and screws (hard water and humidity rust MS); magnetic door stoppers on every door.
- Sunshade (chajja) over every window projects 1'-6" to 2'-0" with a drip groove under the front edge.

Kerala Home, Phase 1

Kerala-style single-storey residence

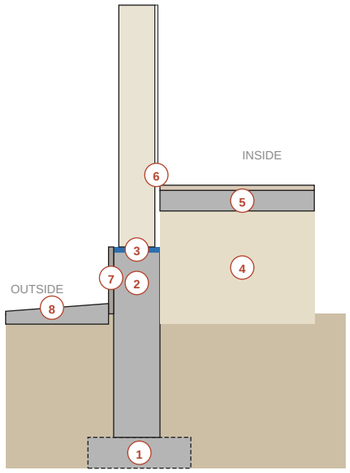
Guntur, Andhra Pradesh

Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

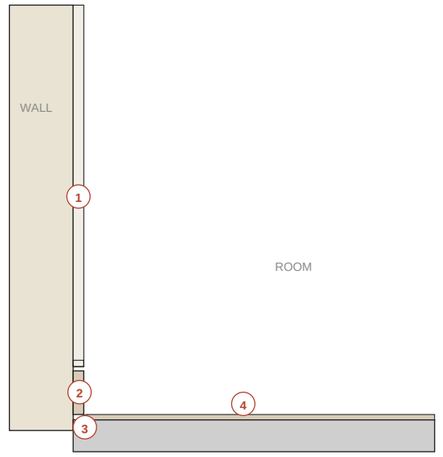
Stage	Draft for professional review
Drawing	**Door and window details**
Sheet	**A-401**
Scale	1:60 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

1 Plinth, DPC, floor and apron



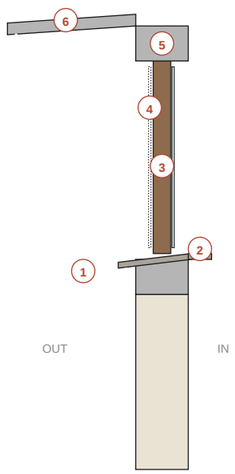
1 Footing by engineer after soil test; on black cotton soil often 1.5-2.5 m deep or under-reamed piles.
2 RCC plinth beam ties all columns; 75 mm sand cushion under it on expansive soil.
3 DPC 40 mm with waterproof compound at plinth top, under the first AAC course.
4 Moorum / gravel fill compacted in 150 mm layers; anti-termite treatment at each layer.
5 PCC bed 100 mm with polythene sheet, then screed and tile.
6 Skirting 4 in, flush with plaster face (plaster stops in a groove): no dust ledge.
7 Stone cladding on the plinth face to 2 ft: rain splash zone stays clean.
8 3 ft PCC apron sloping 1:30 away; polysulphide joint at the wall so water never reaches the foundation.

2 Flush skirting at floor-wall joint



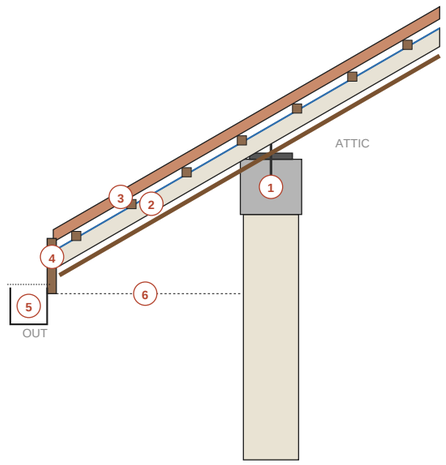
1 Plaster stops 1/4 in above skirting in a clean groove.
2 Skirting tile 4 in high set in the plaster line, face flush with the wall.
3 5 mm gap filled with silicone / PU sealant: allows movement, no cracked tiles.
4 In wet rooms use a cove tile or wall tile down to the floor with epoxy grout instead.

3 Window sill, frame and sunshade



1 External sill slopes out 1:10 and projects 2 in past the plaster with a drip groove underneath.
2 Internal sill in stone, 1/2 in higher than the outer sill (water cannot run inward).
3 Frame on a bed of sealant; gap to wall filled with PU foam and sealed both sides.
4 Insect mesh on a separate openable frame.
5 RCC lintel band continuous around the house at door-head level.
6 Sunshade 1'-6" to 2'-0", top sloped out, drip groove under the front edge (no black streaks on walls).

4 Eaves, gutter and roof build-up



1 RCC ring beam with holding-down bolts cast in; truss shoe bolted, not just welded.
2 Cement-fibre deck + 50 mm insulation + breathable membrane under the battens.
3 Mangalore tiles; eave, verge, hip and ridge tiles clipped or screwed (cyclone wind).
4 Fascia board with drip edge flashing into the gutter.
5 150 mm gutter with leaf guard, falls 1:300 to downpipes.
6 Ventilated soffit with insect / bird mesh: hot attic air escapes at the ridge.

Kerala Home, Phase 1

Kerala-style single-storey residence

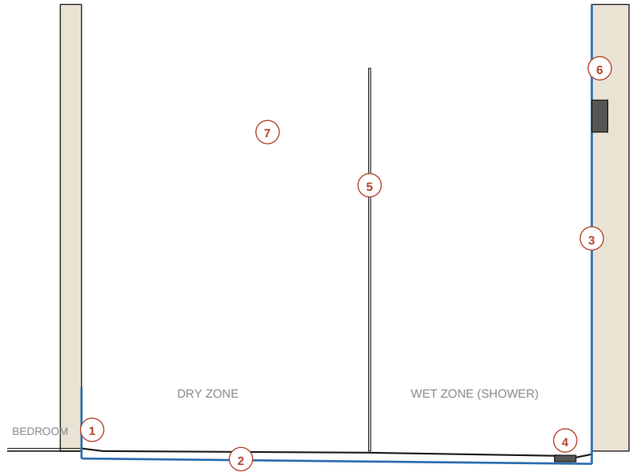
Guntur, Andhra Pradesh

Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

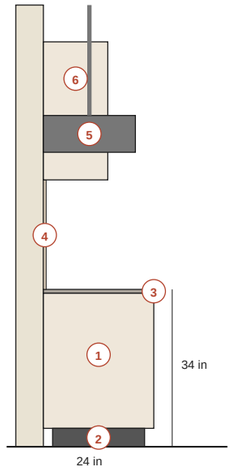
Stage	Draft for professional review
Drawing	**Typical details 1: plinth, floor, window, eaves**
Sheet	**A-501**
Scale	NTS
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

5 Bathroom: wet and dry zones



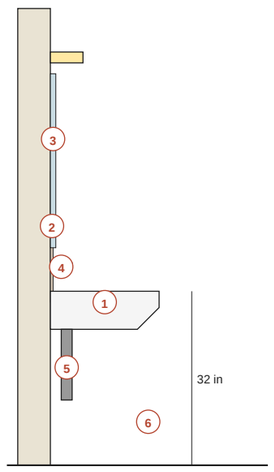
1 Bathroom 12 mm below the bedroom floor with a sloped granite threshold: no raised step to trip on.
2 Waterproof membrane across the floor and 12 in up all walls; 48 h flood test before tiling.
3 Shower walls waterproofed to 7 ft; tiles to ceiling in the shower, 7 ft elsewhere.
4 Linear drain at the far wall; only the wet zone falls (1:60). Dry zone stays dry and safe.
5 8 mm toughened glass partition, 6 in clear of the floor drain line; matt fittings hide hard-water marks.
6 Concealed diverter and pipes in the 7 in wet wall; isolation valve outside the bathroom.
7 Exhaust fan high on the outer wall, opposite the door, ducted outside (not into the attic).

6 Kitchen counter and chimney



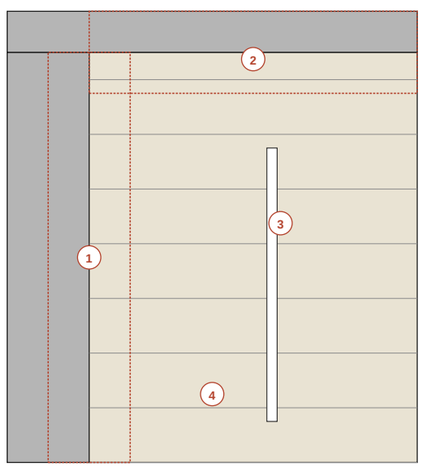
1 Counter top at 34 in (865 mm) for standing cooking; 30-31 in section for a chapati / grinding area.
2 Base units on 4 in legs or a granite plinth: floor can be mopped, no termites, no water damage.
3 20 mm granite (local Chimakurthy / Ongole) with a drip edge; sink under the window, hob on the east run.
4 Stone or tile between counter and wall cabinets, full height; no painted wall behind the hob.
5 Chimney 26-30 in above the hob, ducted straight out through the wall (no long bends).
6 Wall cabinets 14 in deep, starting 22-24 in above the counter; plenty of 16 A sockets at 42 in.

7 Hand-wash basin (serving counter)



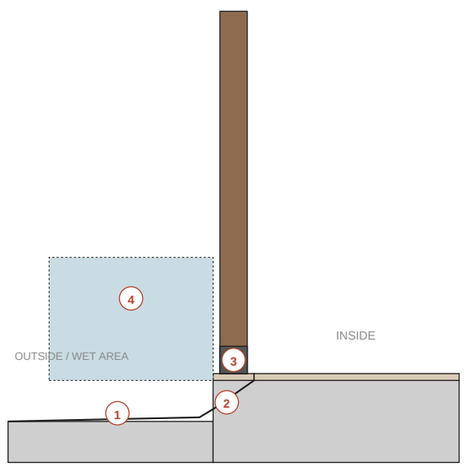
1 Wall-hung basin, rim at 32 in (810 mm); bottle trap, so the floor below can be cleaned.
2 Tiled or stone splash-back 2 ft above the basin: painted walls stain and peel here.
3 Mirror from 40 in to 72 in, light above (not behind) the mirror, socket at 42 in kept 2 ft from the tap.
4 Soap / shelf niche recessed into the wall, sloped forward so water does not pool.
5 Waste pipe into the wall, not across the floor; a floor trap is not needed for a dining-area basin.
6 Children: keep a fold-down step; tap with a long spout so water lands in the bowl, not on the counter.

8 AAC wall joints and chases



1 Galvanised mesh 150 mm wide, centred on every AAC-to-column and AAC-to-beam joint, before plaster.
2 Top course: 20 mm gap under the beam packed with PU foam / flexible filler, not rigid mortar.
3 Chases cut with a wall chaser, max 1/3 of block depth; refill with polymer mortar + mesh.
4 Thin-bed polymer mortar 3 mm, blocks laid dry; first course on the DPC in normal mortar to level.

9 Door thresholds and rain kerbs



1 Outside floor 3-6 in lower than inside and sloping away 1:50: rain never flows in.
2 Granite threshold with a 1 in upstand / weather bar under external doors.
3 Wet-area door frames stand on a granite or WPC block; wood never touches a wet floor.
4 Open wet area: floor 3 in lower and sloping out to the walkway, so rain drains away from the kitchen door.

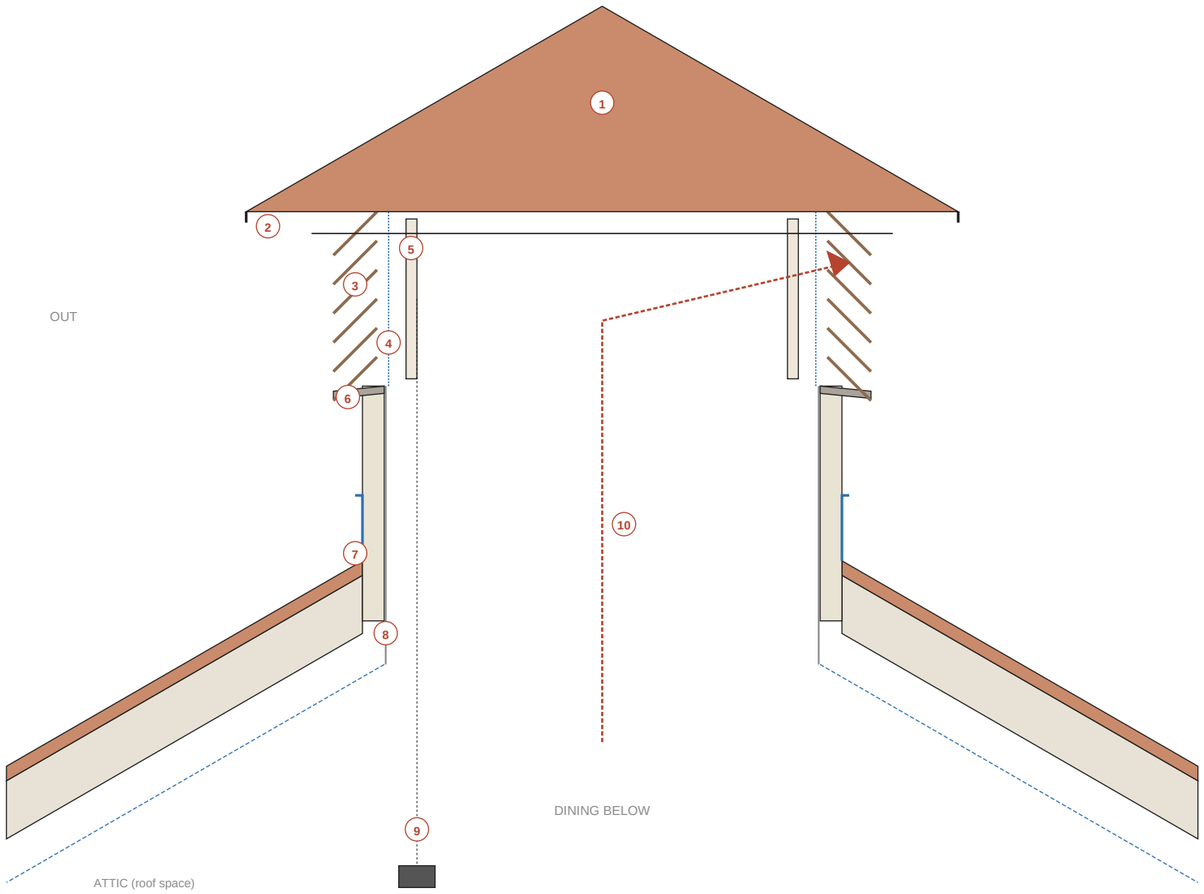
Kerala Home, Phase 1

Kerala-style single-storey residence
Guntur, Andhra Pradesh
Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Typical details 2: bath, kitchen, basin, AAC**
Sheet	**A-502**
Scale	NTS
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

10 Dining ventilator: section



1 Hipped cap at the main roof pitch (30 deg), clay tiles clipped, insulated board under: sheds rain, nothing ponds.
2 Cap overhangs the louvres by 1'-6" with a drip edge, so rain falls well clear of them.
3 Louvre blades at 45 deg sloping down and out with a 4 in lap: no straight line through for wind-driven rain.
4 SS 304 insect mesh about 1.2 mm (18 x 16) in a continuous frame, sealed at every corner: this is what stops mosquitoes.
5 Inner shutters with EPDM rubber seals: closed for AC, cool nights and dust storms. Opening them gives access to clean the mesh.
6 Sill sloped out 1:10 with a drip; any water that gets past the louvres runs back out.
7 9 in minimum upstand above the tiles with lead / membrane flashing and counter-flashing.
8 Shaft lined with insulated fibre-cement board down to the dining ceiling: attic heat and dust stay out.
9 Winding opener (or cord and pulley) with the handle 5 ft above the dining floor; no ladder needed.
10 Hot air from the dining and living rises up the shaft and leaves through the louvres; fresh air enters low through south jalis and windows.

How it works

- Hot air collects under the sloped ceilings of the dining and living. The ventilator is the highest opening in the house, so that air rises out through it.
- Replacement air comes in low and cool through the south jalis, the open wet area and the windows.
- It works best on still, hot evenings, when there is no breeze to cross-ventilate the house.
- No glass, so no direct sun or greenhouse heat on the dining table. Daylight in the dining comes from the living, the nook window and the hall.

Using it

- Summer days and evenings: shutters open.
- AC running in the living: shutters closed, or the cool air leaves through the roof.
- Monsoon storms and cyclone warnings: shutters closed.
- Dust storms or winter nights: shutters closed.
- Twice a year: open the shutters and brush or vacuum the mesh.

Before it is closed up

- Engineer to frame the ventilator into the trusses and check cyclone uplift on the cap.
- Water-test the louvres and flashing with a hose before the dining ceiling is closed.
- Check the mesh frame for gaps with a torch from inside before the shutters are fitted.
- Size: 5 x 5 ft louvre box, 2 ft louvre band, about 20 sq ft of free vent area.

Kerala Home, Phase 1

Kerala-style single-storey residence
Guntur, Andhra Pradesh
Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Typical details 3: dining ventilator**
Sheet	**A-503**
Scale	NTS
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Room schedule

ID	Room	Centreline	Clear	Area sq ft (clear)
R01	Workstation (open, walls optional later)	9'-0" x 8'-0"	8'-7.5" x 7'-7.5"	66
R02	Master bay (old store, open to the bedroom)	9'-0" x 8'-0"	8'-9.5" x 7'-7.5"	67
R03	Dressing alcove	6'-6" x 6'-0"	6'-1.5" x 5'-8.5"	35
R04	Ensuite	6'-6" x 8'-0"	5'-11" x 7'-4"	43
R05	Master bedroom	11'-6" x 14'-0"	11'-1" x 13'-7.5"	151
R07	Bath 2 (attached to Bedroom 2)	5'-6" x 6'-0"	5'-0" x 5'-5"	27
R09	Bedroom 2	15'-0" x 16'-0"	14'-7" x 15'-5"	225
R10	Dining	9'-0" x 8'-0"	8'-9.5" x 8'-0"	70
R19	Dining bay (north window)	9'-0" x 6'-0"	8'-9.5" x 5'-7.5"	49
R11	Pooja	4'-0" x 6'-0"	3'-7" x 5'-5"	19
R13	Wet area (open to the walkway)	6'-0" x 6'-0"	5'-7" x 5'-8.5"	32
R14	Sitting and serving area (open)	18'-0" x 6'-0"	17'-7" x 5'-5"	95
R15	Kitchen	14'-0" x 10'-0"	13'-5" x 9'-3"	124
R16	Living	12'-0" x 12'-0"	11'-7.5" x 11'-3"	131
R17	Poomukham (verandah)	6'-0" x 12'-0"	5'-7.5" x 11'-7.5"	65
	Total (clear)			1200

Door schedule (heights 7'-0"; vents 7'-6" to 9'-0")

ID	Location	Width	Type	Vent above
D01	Main entrance	3'-0"	Teak panelled, double leaf	-
D05	Ensuite	2'-6"	Flush, waterproof (WPC/FRP)	-
D06	Master suite entry, from the sitting area into the master bay	2'-6"	Teak panelled, single leaf	-
D10	Bedroom 2 (from the sitting area)	3'-0"	Flush, teak veneer	-
D15	Bath 2 (attached to Bedroom 2)	2'-6"	Flush, waterproof (WPC/FRP)	-
D11	Pooja	3'-0"	Carved teak, double leaf (opens west; family sits facing east)	Open jali
D12	Kitchen to wet area	2'-6"	Steel or WPC, lockable (the wet area is open to outside)	-
D14	Kitchen entry from dining	3'-0"	Sliding glass, 3 ft (closes off cooking smoke)	-

Window and vent schedule (sill/head above floor)

ID	Room	Wall	Width	Sill	Head	Type
W01	Workstation	S	5'-0"	3'-0"	7'-0"	Timber jali + glazed shutters
W02	Master bay	S	2'-0"	3'-0"	7'-0"	Teak casement, glazed (south light, deep sunshade)
W03	Sitting and serving area	S	5'-0"	3'-0"	7'-0"	Timber jali + glazed shutters
W04	Basin and crockery nook	S	3'-0"	3'-6"	7'-0"	Timber jali + glazed shutters (over the serving counter)
W17	Pooja	S	2'-0"	5'-6"	7'-0"	High vent, jali + mesh (lamp smoke out)
W05	Kitchen	S	3'-0"	3'-6"	7'-0"	Timber jali + glazed shutters
W07	Kitchen	S	5'-0"	3'-6"	7'-0"	Timber jali + glazed shutters
W08	Kitchen	E	6'-0"	3'-6"	7'-0"	Timber jali + glazed shutters
W09	Ensuite	N	3'-0"	5'-0"	7'-0"	Frosted casement
W10	Ensuite	W	4'-0"	5'-0"	7'-0"	High jali vent
W11	Master bedroom	N	6'-0"	3'-0"	7'-0"	Teak casement, glazed
W12	Bedroom 2	N	5'-0"	3'-0"	7'-0"	Teak casement, glazed
W13	Bath 2	N	2'-2.5"	5'-0"	7'-0"	Frosted casement, top-hung vent
W18	Dining bay	N	6'-0"	3'-0"	7'-0"	Teak casement, glazed (north light for the dining)
W14	Living	N	8'-0"	2'-0"	8'-0"	Teak casement, tall, glazed
W16	Basin and crockery nook	S	3'-0"	3'-6"	7'-0"	Timber jali + glazed shutters (over the serving counter)
SK1	Roof	-	-	-	-	Ventilator over dining, 5 x 5 ft, no glass: hipped tiled cap at 30 deg with 1'-6" overhang; 2 ft band of 45 deg louvres on all four sides; SS insect mesh behind; sealed inner shutters on a winding opener; 9 in upstand; lined shaft to the dining ceiling (see A-503)
GV1 -3	Roof	-	-	-	-	Three glazed/jali vents in front gable (mukhappu), morning light into living

Finishes

Space	Floor	Walls / skirting	Ceiling
Living, dining, hall	Red oxide (IPS) or polished Kota / Tandur stone	4 in flush skirting in floor stone; lime-finish or mineral paint	Sloped, follows roof, insulated board under deck
Bedrooms, offices, dressing	Matt vitrified 2x2 ft or red oxide	4 in flush skirting; putty + washable emulsion	False ceiling at 9'-6", 50 mm insulation above
Pooja	White marble or teak	Teak panelling behind idol; 4 in marble skirting	Teak / board at 9'-6"
Kitchen	Anti-skid matt vitrified or Kota, mid-tone	Tile / stone full height between counter and wall cabinets	Moisture-resistant board at 9'-6"
Ensuite, Bath 2	Anti-skid ceramic (R11), falls 1:60 to drain	Tiles to 7 ft; full height in shower; waterproofing 12 in up all walls, 7 ft in shower	Moisture-resistant board / PVC at 8'-6"
Wet area	Rough Cuddapah / Kota stone, falls to drain	4 ft stone or tile dado	Open truss
Family store, sitting area	Red oxide or vitrified	4 in flush skirting; washable paint	Board at 9'-6"
Poomukham, steps	Flamed / leather-finish granite (not polished)	Granite skirting; plinth face clad 2 ft (splash zone)	Exposed rafters + board
Exterior	Plinth apron: PCC 3 ft wide sloped away	AAC + 20 mm plaster with waterproof additive; elastomeric paint	-
Roof (all)	Mangalore clay tiles, every edge tile clipped / screwed	Membrane + 50 mm insulation on cement-fibre deck	Steel truss on RCC ring beam

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

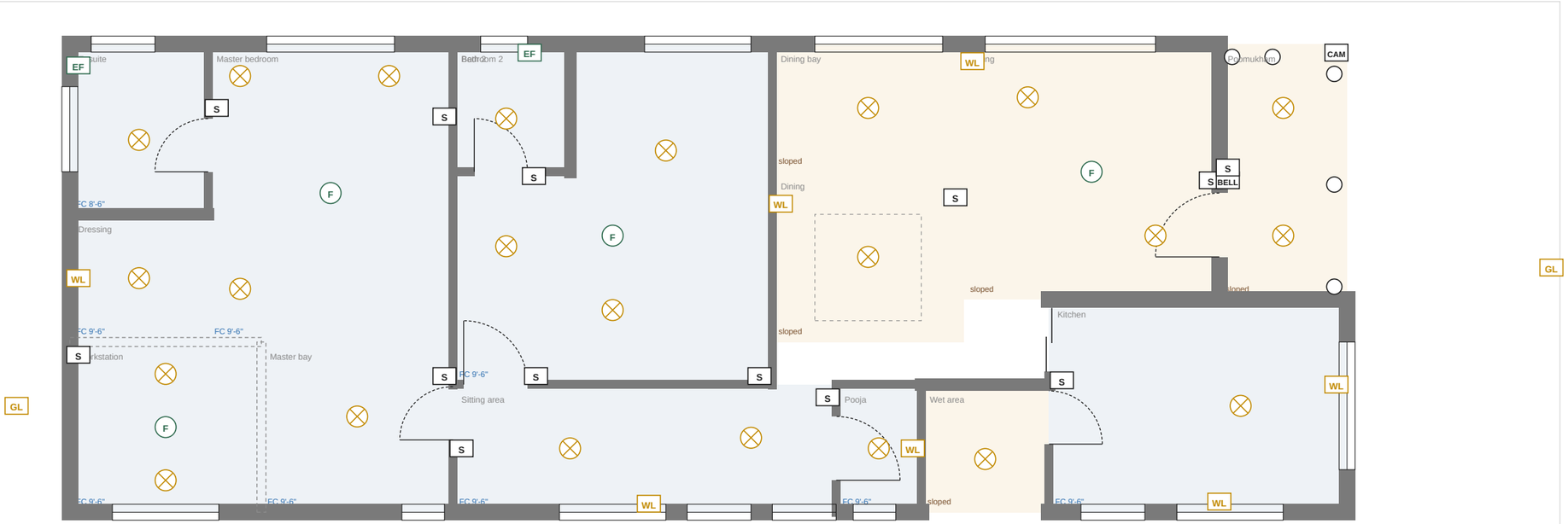
Stage	Draft for professional review
Drawing	**Schedules**
Sheet	**A-601**
Scale	-
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east



Lighting schedule

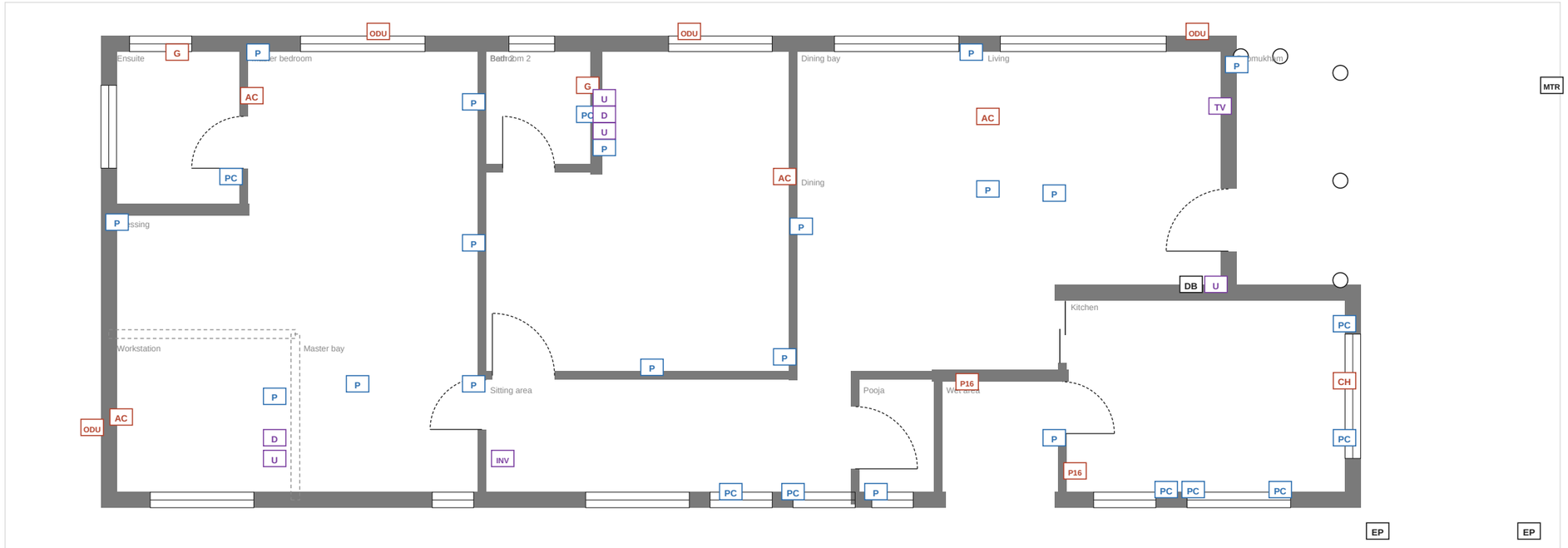
Code	Item	Height	No.
L	Ceiling light	ceiling	23
WL	Wall light / mirror light	6'-6"	7
F	Ceiling fan point	ceiling	4
EF	Exhaust fan	7'-0"	2
S	Switchboard	4'-0"	13
BELL	Bell push	4'-6"	1
CAM	CCTV point	8'-0"	1
GL	Outdoor / gate light	7'-0"	2

Lighting and ceiling notes

- Switchboards on the latch side of every door at 4 ft; 2-way switches at both bedroom beds and both ends of the hall.
- Bathroom switches outside the door; geyser switch with indicator lamp outside too.
- Fans at least 7 ft clear of the floor; long downrods in sloped ceilings; fan hooks cast or bolted, never in false ceiling board.
- Warm 2700-3000 K light in living and bedrooms, 4000 K in kitchen and offices; no downlights directly over the bed head.
- False ceilings (blue) with 50 mm insulation above; access panel near every light cluster and near AC pipe runs.
- Lights, fans, office sockets and router on the inverter circuit; outdoor lights on a timer or photocell.

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Lighting, fans and ceilings**
Sheet	**E-101**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground



Power schedule

Code	Item	Height	No.
CH	Kitchen chimney point (16 A)	7'-0"	1
P	6 A socket	1'-0"	17
PC	6 A socket at counter	3'-6"	9
P16	16 A power socket	varies	2
U	Inverter (UPS) socket	1'-0"	4
D	Data point (Cat6)	1'-0"	2
TV	TV + data point	3'-6"	1
AC	AC indoor unit (16 A socket, sleeve, drain)	7'-6"	4
ODU	AC outdoor unit bracket	7'-0" outside	4
G	Geyser (16 A)	6'-6"	2
DB	Distribution board	5'-0" to top	1
INV	Inverter + battery	floor	1
MTR	Meter at gate	5'-0"	1
EP	Earth pit	ground	2

Power notes

- Separate MCB circuit for each AC, geyser, kitchen power, washing machine and pump; RCCB 30 mA on all socket circuits.
- AC: 16 A socket at 7'-6", 65 mm sleeve sloping out, drain pipe to a gully (not onto the wall); ODUs on shaded north / west walls on brackets.
- AC indoor units never above the bed head or desk chair: cold draught on the head is the common complaint.
- Work desks (Office 2, Bedroom 2): 2 x UPS sockets + data + 1 normal socket at desk height; Cat6 from the router to the TV and both desks.
- Kitchen: 5-6 counter sockets at 42 in, 16 A for the fridge, chimney and microwave / OTG; none within 2 ft of the sink.
- DB in the living wall near the entrance with 25% spare ways; meter at the gate; 2 earth pits; surge protector.
- Empty 25 mm conduits from the roof (solar PV), gate (CCTV, bell, intercom) and sump (pump controller) to the DB.

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

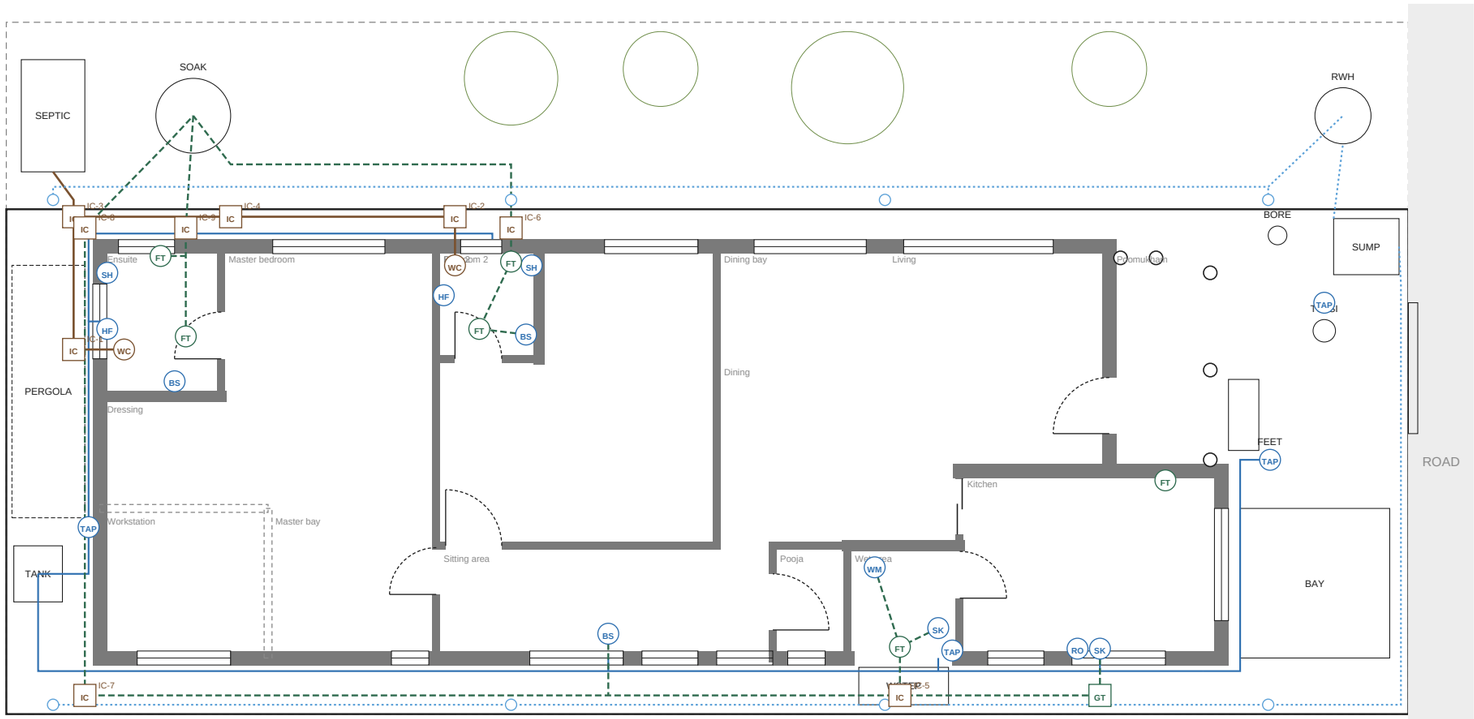
Stage	Draft for professional review
Drawing	**Power, AC and data**
Sheet	**E-102**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east



Water supply (CPVC), from overhead tank

Soil line (uPVC 110 mm) to septic tank

Grey water (uPVC 75 mm) to soak pit

Rainwater to sump filter / recharge pit

Fixtures and points

Code	Item	No.
WC	Wall-hung WC, concealed cistern	2
HF	Health faucet	2
BS	Wash basin	3
SH	Shower mixer (hot + cold)	2
FT	Floor trap / linear drain	6
SK	Sink	2
WM	Washing-machine inlet + drain	1
RO	RO purifier point	1
TAP	Tap	4
IC	Inspection chamber	9
GT	Grease trap	1

Plumbing notes

- Overhead tank feeds by gravity; isolation valve for each bathroom and the kitchen, outside the room.
- Supply pipes run on the outside of walls in clamps or shallow chases, never under the floor slab.
- Soil and grey water kept separate; inspection chamber at every bend and junction; 1:40 falls on 110 mm, 1:30 on 75 mm.
- Grease trap before the kitchen line joins anything; vent pipe from each WC stack above the eaves.
- Soak pit and septic in the north strip, 6 ft+ from walls: wet black cotton soil swells and cracks the house.
- Rain from both gutters to the sump through a first-flush filter; overflow to a recharge pit.
- Test the bore / supply water: RO for drinking, softener for bathrooms if hardness is high; matt or PVD fittings.

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Water supply and drainage**
Sheet	**P-101**
Scale	1:100 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground